

Debopriyo Roy

Senior Software Developer – Cloud Data Platforms & Infrastructure | AWS | Terraform IaC | Apache Spark | EMR | Python | Scalable Data Pipelines

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About Me

Cloud Platform and Data Engineer with 3+ years of experience designing scalable, cloud-native data platforms on AWS, specializing in Infrastructure as Code using Terraform and distributed data processing with Apache Spark. Experience in building secure, automated cloud infrastructure including VPC networking, IAM policies, and production-grade data pipelines to process large-scale structured and unstructured datasets. Experienced in developing high-performance data engineering workflows on Amazon EMR and S3, supporting advanced analytics and machine learning pipelines. Strong Understanding in Python development, cloud infrastructure automation, and resilient data platform design, enabling reliable, scalable, and cost-efficient data processing systems for enterprise analytics and AI-driven applications.

Education

Memorial university Of Newfoundland

Master's In Applied Science (Masc.): Computer Engineering. Cumulative Average: 76.2%

St John's, NL

Jan 2023-Sept 2024

Calcutta University (ALL INDIA GATE RANK: 583)

Master's In Technology in VLSI Design and Embedded System (Drop-out)

West Bengal, India

Sept 2021- Dec 2022

West Bengal University of Technology

Bachelor's In Electronics & Communication Engineering. Cumulative Average: 75.2%

West Bengal, India

May 2012-May2016

Work Experience

Software Engineer (Volunteer) | Centre for Analytics, Informatics and Research (CAIR) @ MUN.

St John's, NL

Talent Opportunity Hub:

July 2025 – Present

- Built and deployed an AI-powered job-matching platform serving students and employers, improving onboarding efficiency through automated workflows.
- Designed multi-step onboarding with verification and admin approvals, reducing manual review effort by almost ~80% and improving data integrity.
- Implemented CI/CD pipelines with Docker/Podman and GitHub Actions, enabling consistent, repeatable deployments.
- Collaborated cross-functionally with academic stakeholders to translate requirements into production-ready cloud services.

Co-Owner | Multi-sector Family Business

west Bengal, India

2011-2022

- Co-owner and primary decision-maker in a decade-old, high-volume family business serving **1,000+ daily customers** across gas stations, hotels, and heavy vehicle operations. Developed strong business acumen and cultivated advanced communication and customer-facing skills through hands-on business involvement beginning at age 19.
- Gained the ability to stay calm under pressure, resolve issues quickly, and ensure customers felt supported. Worked my way up through the business, from assisting with daily workflows and coordination to overseeing operations, before moving to Canada to pursue my master's degree.
- Remain actively involved in strategy, growth planning, and key operational decisions while co-owners oversee day-to-day operations.

Customer Service Associate: Front Desk Staff | Loblaw's Inc.

St John's, NL

January, 2023 – Present

- Served as the initial point of contact for over **100 daily customers**, clearly communicating services and while identifying needs and resolving concerns.
- Handled objections and escalations professionally, using active listening and empathy to provide a pleasant customer experience to a varied audience.
- Managed front-desk activities, such as scheduling, check-ins, and visitor logs, to ensure correct documentation and efficient workflow execution.
- Collaborate with leadership to address problems while maintaining high performance in a fast paced workplace with agility and multitasking.

Bengal College of Polytechnic

West Bengal, India

Assistant Professor.

Jan 2020-July 2020

- Taught Analog and Digital Electronics and Control Systems Engineering to strengthen students' core concepts.
- Developed and delivered engaging lectures, lab sessions, and assessments to connect theoretical learning with real-world applications.
- Guided and mentored students in academic projects, encouraging critical thinking, creativity, and problem-solving in the field of electronics and control engineering.

ASIC Design Engineer | Technical & Client-Facing Liaison | EMSIG Technologies Pvt Ltd.

Mumbai, India

August 2016-December 2019

- Partnered with defense and biomedical clients to translate complex technical requirements into clear system specifications and delivery plans.
- Supported pre-sales and solution design by producing technical documentation, solution briefs, and client-facing demos.
- Acted as a bridge between engineering teams and stakeholders, improving delivery clarity and long-term client engagement.

Skills

Cloud Platforms	AWS(EC2,ECS,EKS,ELB,Lambda,RDS,S3,CloudFront,IAM,CloudWatch,EventBridge,CloudFormation, API Gateway),EMR,AWS Step Function.
Automation & Unix	Terraform, Ansible,Docker, Kubernetes, DevContainer, CI/CD (GitHub Actions, Jenkins, CodePipeline, CodeBuild), Linux.
Database Administration	Postgres SQL,DynamoDb, Glue, Athena, SQLite, , Redshift Spectrum,ETL pipelines,Data Ingestion, Data Normalization.
Programming languages	Python, Java, SQL, Bash/Shell Scripting, REST APIs, YAML, JSON.
Monitoring, Observability	AWS X-Ray, Cloudwatch metrics & Logs, Prometheus, Grafana.
Web Frameworks & Backend	Spring Boot, Django, HTMX, Bootstrap 5, JavaScript,LangChain,LangGraph.
Data Processing Pipelines	Apache Spark, PySpark, Amazon EMR, Data Pipelines, ETL Pipelines, Data Lake Architecture Data Processing, Structured & Unstructured Data Processing.
Machine learning platforms	Amazon SageMaker, SageMaker Pipelines, PyTorch, TensorFlow,EMR, XGBoost, Amazon Bedrock, Model training, deployment & ML workflow automation.
Distributed System & Security	Scalable system design, Fault tolerance, IAM least privilege, Secure Cloud Architectures.

Projects

1 .AWS Infrastructure Automation with Terraform CLI & Terraform Cloud.

Description: Designed and automated enterprise-grade AWS infrastructure using Terraform Infrastructure as Code, provisioning core services including VPC networking, IAM roles and policies, EC2 compute, storage, and monitoring components through reusable Terraform modules and Terraform Cloud workspaces. Implemented remote state management using Amazon S3 with DynamoDB locking to enable secure collaborative deployments and prevent configuration drift. Automated network connectivity, IAM access models, and infrastructure provisioning pipelines, improving deployment consistency, governance, and scalability while reducing manual environment setup time by ~70%.

Tools Used: AWS (VPC, Subnets, EC2, S3, IAM, Lambda, Glue, CloudWatch), Terraform CLI, Terraform Cloud, Terraform Modules, Infrastructure as Code (IaC), Sentinel Policy-as-Code, Remote State (S3 + DynamoDB).

2. End-to-End CI/CD & observability platform for Microservices (AWS).

Description: Built production-grade CI/CD pipelines using Jenkins and Terraform to automate build, test, security scanning, and deployment of containerized microservices on AWS (EKS/ECR/EC2). Integrated SonarQube quality gates, CloudWatch logging, and Prometheus/Grafana monitoring to improve deployment reliability, enforce standards, and reduce manual release effort by ~80%.

Tools Used: AWS (EKS, ECS, EC2, ECR, S3, IAM, CloudWatch), Terraform (IaC), Jenkins, Docker, Git/GitHub, Maven, SonarQube, Prometheus, Grafana.

3. Event-driven AWS Serverless ETL Pipelines.

Description: Developed an event-driven serverless ETL pipeline where S3 uploads triggered AWS Lambda to clean, normalize, and transform JSON into Parquet for scalable analytics. Automated schema discovery using AWS Glue Crawlers and enabled fast querying through Athena, with CloudWatch dashboards and alarms for monitoring pipeline health and reliability.

Tool Used: AWS S3, AWS Lambda, AWS Glue (Crawlers), AWS Athena, CloudWatch,Parquet,Json.

4. Enterprise Credit Card Fraud Detection Platform.

Description: Designed and implemented a scalable data processing platform for fraud detection, analyzing 300,000+ financial transactions using distributed data engineering and machine learning workflows. Built Spark-based ETL pipelines on Amazon EMR to process large-scale transaction data stored in Amazon S3, performing feature engineering and transformation for downstream analytics and model training. Developed reliable data pipelines to convert raw transaction datasets into curated Parquet datasets for high-performance analytics and machine learning in Amazon SageMaker. Implemented distributed processing using Apache Spark (PySpark) and optimized data pipelines for large-scale financial datasets.

Tools Used: AWS EMR (Spark, PySpark), Amazon S3 Data Lake, EMR Studio, SageMaker Studio, IAM Roles & Policies, VPC Networking , Python XGBoost, Scikit-learn, SMOTE (imbalanced-learn), SHAP Explainability.

5. AI-Powered Customer Support Agent using Amazon Bedrock AgentCore.

Description: Built a production ready AI customer support agent using Amazon Bedrock AgentCore enabling automated query handling, intent-based response & tool-driven action. Operationalize the workflow with logging,IAM guardrails & cloudWatch monitoring for production readiness.

Tools Used: Amazon Bedrock,Bedrock AgentCore,LangChain,LangGraph,Python,IAM Policies,Cloudwatch.

6. Personal Portfolio Website (debopriyo-roy.in).

Description: Built a responsive personal portfolio website (debopriyo-roy.in) using HTML, CSS, JavaScript, and PHP (Composer) to showcase my profile and projects. Implemented a clean, mobile-friendly UI with a functional contact form to enable direct communication and improve professional visibility.

Tools used: HTML, CSS, PHP,Composer,Javascript.

Certifications

[AWS Certified Solutions Architect – Associate.](#) | [HashiCorp Certified: Terraform Associate \(003\).](#)

[Terraform Associate \(003\) — HashiCorp Certification,Udemy](#) | [AWS Machine Learning Engineer Associate, Udemy](#)